Boundary values by comparison

The theorem

Theorem (the Dirichlet problem on a union of squares)

Let Ω be a finite union of grid squares and g continuous on $\partial\Omega$. There is exactly one $u \in C^2(\Omega) \cap C(\overline{\Omega})$ with $\Delta u = 0$ in Ω and $u = g$ on $\partial\Omega$.

The proof has four steps.

1. **Reduction** to g linear on each boundary edge of the coarsest grid.
2. **A candidate.** For such g, the finite element solutions U_n converge in energy to a function u (Lecture 3).
3. **Harmonic, and unique.** u is smooth and harmonic in Ω (Weyl's lemma); a harmonic function in $C(\overline{\Omega})$ is determined by its boundary values (the maximum principle). Both Lecture 4.
4. **Boundary values.** $u(z) \to g(p)$ as $z \to p \in \partial\Omega$.

Today: step 4, by **energy** and a **comparison** of u with explicit harmonic functions. Steps 1 to 3 are used as they stand.

Step 1: reduction to piecewise linear boundary values

Suppose the theorem holds for boundary data linear on each boundary edge of some grid, and let g be continuous.

Linearise at level k. Let g_k equal g at the boundary nodes of level k and be linear on each boundary edge of level k. Since g is uniformly continuous, $g_k \to g$ uniformly on $\partial\Omega$.

Solve for g_k. By assumption there is u_k, harmonic in Ω and continuous on $\overline{\Omega}$, with $u_k = g_k$ on $\partial\Omega$ (steps 2 to 4, with the grid of level k as coarsest grid).

Let $k \to \infty$, by the maximum principle. $u_k - u_j$ is harmonic and continuous on $\overline{\Omega}$, so by the stability corollary (Lecture 4)

$$\max_{\overline{\Omega}} |u_k - u_j| \;=\; \max_{\partial\Omega} |g_k - g_j| \;\longrightarrow\; 0 \qquad (j, k \to \infty).$$

So $u_k \to u$ **uniformly** on $\overline{\Omega}$: u is continuous on $\overline{\Omega}$, equals g on $\partial\Omega$, and is harmonic in Ω as a uniform limit of harmonic functions (Lecture 4). $\square$

Steps 2 and 3: what we may use

From now on g is linear on each boundary edge of the coarsest grid, and U_n is the finite element solution on the grid of size $\varepsilon_n = 2^{-n}\varepsilon_0$: equal to g at the boundary nodes, bilinear on each square. Points of the plane are $z = x + iy$.

- $U_n = g$ on $\partial\Omega$: on a boundary edge U_n is linear and agrees with g at both ends.
- $U_n \to u$ and $\nabla U_n \to \nabla u$ in $L^2(\Omega)$ (Lecture 3).
- u is smooth and harmonic in Ω (Weyl's lemma, Lecture 4).
- The maximum principle for harmonic functions in $C^2(\Omega) \cap C(\overline{\Omega})$ (Lecture 4).

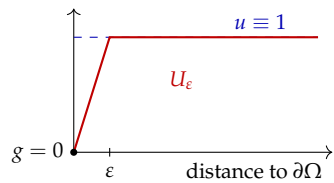
To be proved today (step 4): for every $p \in \partial\Omega$, $u(z) \to g(p)$ as $z \to p$. Then u, extended by g on $\partial\Omega$, is the function of the theorem.

Where the boundary values hide

We know: $U_n = g$ on $\partial\Omega$ for every n, and $U_n \to u$ in L^2, with u harmonic. **We want:** $u = g$ on $\partial\Omega$.
This does not follow, and no argument using only these facts can prove it. Take $g \equiv 0$ and

$$U_\varepsilon(z) := \min\big(1,\ \operatorname{dist}(z, \partial\Omega)/\varepsilon\big).$$

Then $U_\varepsilon = 0$ on $\partial\Omega$, and $U_\varepsilon \to 1$ in L^2: it differs from 1 only on a layer of width ε. The limit $u \equiv 1$ is harmonic, with the wrong boundary values.



The example has unbounded energy: $|\nabla U_\varepsilon| = 1/\varepsilon$ on the layer, so $\iint |\nabla U_\varepsilon|^2\, dA \approx |\partial\Omega|/\varepsilon \to \infty$. Our U_n do not: ∇U_n converges in L^2.

L^2 limits forget the boundary values. Energy limits remember them.

The plan

1. **A comparison lemma**, for our u: the harmonic L^2 limit of the U_n, with $\nabla U_n \to \nabla u$ and $U_n = g$ on $\partial\Omega$. If v is continuous on $\overline{\Omega}$, harmonic in Ω, and $v \geq g$ on $\partial\Omega$, then $v \geq u$: a maximum principle for $v - u$ whose boundary inequality is checked against g, the boundary values of the U_n, since those of u are not yet known to exist.
2. **Barriers.** At each $p \in \partial\Omega$, explicit harmonic functions $w \geq 0$: of size about 1 at p, but as large as we like on the rest of $\partial\Omega$. Then $v_\pm := g(p) \pm (\text{small} + k\,w)$ squeeze u at p.

The lemma is where the energy works. The barrier is a logarithm: the potential of a point charge placed just outside Ω.

The comparison lemma

Lemma (comparison)

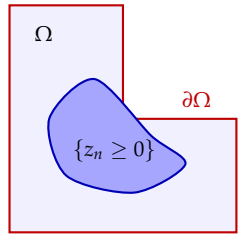
Let u be the function of Lecture 3: harmonic in Ω, and the L^2 limit of the U_n, with $\nabla U_n \to \nabla u$ in L^2 and $U_n = g$ on $\partial\Omega$. Let v be continuous on $\overline{\Omega}$ and harmonic in Ω. If $v \geq g$ on $\partial\Omega$, then $v \geq u$ in Ω. If $v \leq g$ on $\partial\Omega$, then $v \leq u$ in Ω.

Proof. Fix $\sigma > 0$, and put

$$z_n := U_n - \sigma - v, \qquad z := u - \sigma - v.$$

We show $z \leq 0$ in Ω; then $u \leq v + \sigma$, and $\sigma \downarrow 0$.

Everything rests on the positive part $z_n^+ := \max(z_n, 0)$: it vanishes near $\partial\Omega$, its energy tends to zero, and it tends to z^+. First a list of what we need about $t \mapsto t^+$.



On $\partial\Omega$, $z_n = g - \sigma - v \leq -\sigma$: the set $\{z_n \geq 0\}$ stays inside Ω.

What we need about $t \mapsto t^+$

Let f be continuous on $\overline{\Omega}$, $f \le -\sigma$ on $\partial\Omega$ for some $\sigma > 0$, and smooth on each closed square of some grid inside Ω (in mind: $f = z_n = U_n - \sigma - v$). Write $f^+ := \max(f, 0)$.

- **(a) Support.** $f^+ = 0$ outside the compact set $K := \{f \ge 0\} \subset \Omega$.
- **(b) Contraction.** $|a^+ - b^+| \le |a - b|$ for real a, b.
- **(c) Gradient.** $\nabla f^+ = \nabla f$ where $f > 0$, $\nabla f^+ = 0$ where $f \le 0$, almost everywhere; so $\nabla f \cdot \nabla f^+ = |\nabla f^+|^2$. On any line f is piecewise C^1 near K, so f^+ is Lipschitz there, and $(f^+)' = f'$ where $f > 0$, 0 where $f < 0$, and 0 on $\{f = 0\}$ wherever it exists (a minimum of f^+).
- **(d) Green.** $\int\int_\Omega \nabla z \cdot \nabla f^+ \, dA = -\int\int_\Omega f^+ \Delta z \, dA$ for z smooth in Ω, so $= 0$ if z is harmonic. Integrate by parts along each line; f^+ vanishes near the ends of each interval of Ω on the line, so there are no boundary terms. (Step 1 of Lecture 3 with $w = f^+$.)
- **(e) Poincaré.** $\|f^+\|_{L^2} \le L \, \|\nabla f^+\|_{L^2}$, L the width of Ω: the proof of Lecture 3 only integrated along lines.

Proof of the lemma: the energy of z_n^+ vanishes

Step 1: z_n^+ lives inside Ω. [Uses $U_n = g \le v$ on $\partial\Omega$.] On $\partial\Omega$, $z_n = g - \sigma - v \le -\sigma$. Also z_n is continuous on $\overline{\Omega}$ and, inside Ω, smooth on each closed square of the level-n grid (U_n is bilinear there, v is smooth). So (a)–(e) of slide 8 apply to $f = z_n$.

Step 2: its energy tends to zero. [Uses u harmonic, and $\nabla U_n \to \nabla u$.] By (c), then $z_n - z = U_n - u$, then (d) for the harmonic function z:

$$\begin{aligned}
\iint_\Omega |\nabla z_n^+|^2 \, dA &= \iint_\Omega \nabla z_n \cdot \nabla z_n^+ \, dA \\
&= \iint_\Omega \nabla(U_n - u) \cdot \nabla z_n^+ \, dA + \underbrace{\iint_\Omega \nabla z \cdot \nabla z_n^+ \, dA}_{=0} \\
&\le \|\nabla(U_n - u)\|_{L^2} \, \|\nabla z_n^+\|_{L^2}.
\end{aligned}$$

So $\|\nabla z_n^+\|_{L^2} \le \|\nabla(U_n - u)\|_{L^2} \to 0$, and by (e), $\|z_n^+\|_{L^2} \to 0$.

Proof of the lemma: identify the limit

Step 3: $z_n^+ \to z^+$ **in** L^2**.** [Uses $U_n \to u$ in L^2.] By (b),

$$|z_n^+ - z^+| \;\le\; |z_n - z| \;=\; |U_n - u| \qquad \text{at every point of } \Omega,$$

so $\|z_n^+ - z^+\|_{L^2} \le \|U_n - u\|_{L^2} \to 0$. Together with Step 2, uniqueness of L^2 limits gives $z^+ = 0$ almost everywhere in Ω.

Step 4: almost everywhere to everywhere. z is continuous in Ω, and a continuous function positive at a point is positive on a disk of positive area. So $z \le 0$ in Ω, that is, $u \le v + \sigma$. Let $\sigma \downarrow 0$.

For the mirror statement take $z_n := v - \sigma - U_n$ and $z := v - \sigma - u$; the same steps apply. □

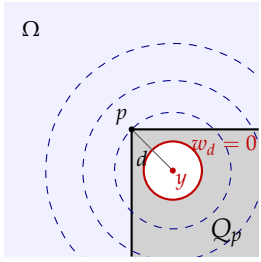
Boundary inequality: z_n^+ **lives inside** Ω**. Gradient convergence: its energy vanishes.**
L^2 **convergence: its limit is** z^+**. Harmonicity of** u**: no cross term.**

The barrier: a point charge outside Ω

Fix $p \in \partial\Omega$. A closed coarse square S containing p is not a square of Ω (else p is interior), so S misses Ω and contains a square Q_p of side $\varepsilon_0/2$ with corner p.
Let y be on the diagonal of Q_p at distance $d \le \varepsilon_0/4$ from p, and

$$w_d(z) := \log \frac{|z-y|}{d/2},$$

the potential of a point charge at y (Lecture 2).



Small at p, large everywhere else as $d \to 0$.

- w_d is harmonic in Ω and smooth on $\overline{\Omega}$, since $y \notin \overline{\Omega}$.
- $w_d \ge 0$ on $\overline{\Omega}$: the disk $|z-y| < d/2$, where $w_d < 0$, lies inside Q_p.
- $w_d(p) = \log 2$.
- $w_d(z) \ge \log(|z-p|/d)$ if $|z-p| \ge 2d$, as $|z-y| \ge |z-p| - d \ge \frac{1}{2}|z-p|$.

Squeezing u at p

Fix $\rho \le \varepsilon_0/2$ and $d \le \rho/2$, let
$\operatorname{osc}_\rho := \max\{|g(q) - g(p)| : q \in \partial\Omega,\ |q-p| \le \rho\}$, and put

$$v_\pm := g(p) \pm \big(\operatorname{osc}_\rho + k\, w_d\big), \qquad k := \frac{2\max|g|}{\log(\rho/d)}.$$

On $\partial\Omega$, $v_- \le g \le v_+$**.** Within ρ of p, $v_+ \ge g(p) + \operatorname{osc}_\rho \ge g$, since $w_d \ge 0$.
Farther than ρ from p, $w_d \ge \log(\rho/d)$, so $k\, w_d \ge 2\max|g|$ and $v_+ \ge g$.
Likewise $v_- \le g$.

So, by the lemma, $v_- \le u \le v_+$ in Ω, i.e. $|u(z) - g(p)| \le \operatorname{osc}_\rho + k\, w_d(z)$. Let $z \to p$; since $w_d(p) = \log 2$,

$$\limsup_{z\to p} |u(z) - g(p)| \ \le\ \operatorname{osc}_\rho + \frac{2\max|g|\log 2}{\log(\rho/d)}.$$

The left side involves neither d nor ρ: let $d \to 0$, then $\rho \to 0$. So $u(z) \to g(p)$:
step 4 is proved. □

What we have proved

Step 4. For g linear on the coarse boundary edges, the energy limit u of Lecture 3 satisfies $u(z) \to g(p)$ as $z \to p$, for every $p \in \partial\Omega$. With steps 1 to 3: the Dirichlet problem on a union of squares has exactly one solution for every continuous g.

What went in. Lecture 3: the two limits $U_n \to u$, $\nabla U_n \to \nabla u$, and $U_n = g$ on $\partial\Omega$. Lecture 4: Weyl's lemma and the maximum principle. Today: integration by parts against z_n^+, and the logarithmic barrier.

Beyond grids. The lemma used nothing about the U_n except the three properties above:

A harmonic function that is an energy limit of functions with boundary values $\leq v$ on $\partial\Omega$ is $\leq v$ in Ω.

With "H^1" and "trace" put in, this is the weak maximum principle of the modern theory of elliptic equations.